

0.1 Schur's Lemma

0.1.1 Triangularization Lemma

Lemma 0.1.1. Let V be a finite-dimensional vector space over F .
If the characteristic polynomial of a linear operator $T: V \rightarrow V$ splits over F ,
then there exists a basis β s.t. $[T]_\beta$ is an upper-triangular matrix.

Proof. Put $n = \dim V$. Using mathematical induction for n . If $n = 1$, trivial.

Suppose that for $n > 1$, the statement holds for every space having dimension $n - 1$.

Let V be a vector space with $\dim V = n$, and let $T: V \rightarrow V$ be a linear operator.

Since the characteristic polynomial of T splits over F , there exists an eigenvalue $\lambda_1 \in F$ corresponding to $v_1 \in V$.

By the Replacement Thm, we obtain a basis for V containing v_1 :

$$\{v_1, v_2, \dots, v_n\}.$$

Denote $W_1 = \langle v_1 \rangle, W_2 = \langle v_2, \dots, v_n \rangle$. Clearly, $V = W_1 \oplus W_2$.

Observe that W_1 is T -invariant but W_2 is not. Define two projections

$$P_1: V \rightarrow V: x_1 + x_2 \mapsto x_1 \quad \text{and} \quad P_2: V \rightarrow V: x_1 + x_2 \mapsto x_2, \quad (x_1 \in W_1, x_2 \in W_2)$$

Then, for any $x \in W_2$, $(P_2 \circ T)(x) \in W_2$, thus we can consider a restriction

$$(P_2 \circ T)|_{W_2}: W_2 \rightarrow W_2; x \mapsto (P_2 \circ T)(x).$$

By the assumption, there exists a basis $\{w_2, \dots, w_n\}$ for W_2 s.t. for $2 \leq j \leq n$,

$$(P_2 \circ T)(w_j) = \sum_{i=2}^j a_i w_i \quad \text{for some } a_i \in F.$$

Now, define for $1 \leq k \leq n-1$, $u_{k+1} = v_1 + w_{k+1}$. Then, $\{v_1, u_2, \dots, u_n\}$ is a basis for V , and satisfies for all $1 \leq j \leq n$,

$$\begin{aligned} T(v_1) &= \lambda_1 v_1 \\ T(u_2) &= \lambda_1 v_1 + a_2 w_2 \\ T(u_3) &= \lambda_1 v_1 + a_2 w_2 + a_3 w_3 \\ &\vdots \\ T(u_n) &= \lambda_1 v_1 + a_2 w_2 + \dots + a_n w_n. \end{aligned}$$

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